Microsoft 365 PowerShell Modules - Install, Load and Connect

Quick reference for administrators. Commands are for PowerShell 7 (recommended) or Windows PowerShell 5.1. Run as your normal user: **-Scope CurrentUser** needs no admin rights. Prepared November 2026.

1. Prerequisites (once per computer)

Install PowerShell 7 and check the version

```
winget install --id Microsoft.PowerShell --source winget     # Windows
brew install --cask powershell                               # macOS (Homebrew)
$PSVersionTable.PSVersion                                    # 7.4 or later recommended
```

Windows PowerShell 5.1 only: TLS 1.2 and the NuGet provider

```
[Net.ServicePointManager]::SecurityProtocol =
    [Net.ServicePointManager]::SecurityProtocol -bor [Net.SecurityProtocolType]::Tls12
Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Scope CurrentUser -Force
```

Allow locally run and signed scripts; make sure the PowerShell Gallery is registered

```
Set-ExecutionPolicy RemoteSigned -Scope CurrentUser
Get-PSRepository                                 # should list PSGallery
Register-PSRepository -Default                   # only if PSGallery is missing
Unblock-File .\Initialize-M365AdminModules.ps1   # only for scripts downloaded from the web
```

2. Which modules you need

Module	Use it for	Connect with
Microsoft.Graph.Authentication	Sign-in to Graph; Invoke-MgGraphRequest for any Graph query	Connect-MgGraph
Microsoft.Graph.Users / Users.Actions	Users, managers, licence assignment, revoke sessions	(Graph)
Microsoft.Graph.Groups	Groups and membership	(Graph)
Microsoft.Graph.Identity.DirectoryManagement	Roles, domains, subscribed SKUs, devices	(Graph)
Microsoft.Graph.Identity.SignIns	Conditional Access, authentication methods	(Graph)
Microsoft.Graph.Reports	Sign-in and audit logs, usage reports	(Graph)
ExchangeOnlineManagement	Exchange Online; Security & Compliance (Purview) PowerShell	Connect-ExchangeOnline / Connect-IPPSSession
MicrosoftTeams	Teams policies, voice, meetings	Connect-MicrosoftTeams
PnP.PowerShell (PS 7)	SharePoint Online and OneDrive	Connect-PnPOnline
Microsoft.Online.SharePoint.PowerShell	SharePoint Online admin (classic module)	Connect-SPOService
Microsoft.Entra	Entra ID cmdlets with AzureAD-style names (built on Graph)	Connect-Entra
ImportExcel	Export results to .xlsx without Excel installed	-

Microsoft.Graph is a meta-module of about 40 sub-modules. Install it whole if you like, but import only the sub-modules you need: importing everything is slow, and in Windows PowerShell 5.1 it can hit the 4,096-function limit.

3. Install (once) and update

Core set: Graph for users/groups/licences/reports, Exchange Online, Excel export

```
$core = 'Microsoft.Graph.Authentication', 'Microsoft.Graph.Users',
        'Microsoft.Graph.Users.Actions', 'Microsoft.Graph.Groups',
        'Microsoft.Graph.Identity.DirectoryManagement', 'Microsoft.Graph.Identity.SignIns',
        'Microsoft.Graph.Reports', 'ExchangeOnlineManagement', 'ImportExcel'
Install-Module -Name $core -Scope CurrentUser -Repository PSGallery -AllowClobber -Force
```

Optional: Teams, SharePoint, Entra, extra Graph areas

```
Install-Module MicrosoftTeams, Microsoft.Entra -Scope CurrentUser -Force
Install-Module PnP.PowerShell -Scope CurrentUser -Force                  # PowerShell 7
Install-Module Microsoft.Online.SharePoint.PowerShell -Scope CurrentUser # 5.1 alternative
Install-Module Microsoft.Graph.Applications, Microsoft.Graph.DeviceManagement,
    Microsoft.Graph.Security, Microsoft.Graph.Identity.Governance -Scope CurrentUser -Force
Install-Module Microsoft.Graph -Scope CurrentUser -Force          # or: the whole Graph SDK
Install-Module Microsoft.Graph.Beta -Scope CurrentUser -Force     # preview APIs (optional)
```

Update everything, and keep all Graph sub-modules on the same version

```
Update-Module -Name Microsoft.Graph*, ExchangeOnlineManagement, MicrosoftTeams, PnP.PowerShell
Get-InstalledModule Microsoft.Graph* | Sort-Object Version | Format-Table Name, Version
```

4. Load the modules (each new PowerShell session)

Load Microsoft.Graph.Authentication first: it brings in the sign-in libraries the other Graph sub-modules share. PowerShell also auto-loads a module the first time you use one of its commands, but explicit imports make version problems show up immediately.

```
Import-Module Microsoft.Graph.Authentication
Import-Module Microsoft.Graph.Users, Microsoft.Graph.Users.Actions, Microsoft.Graph.Groups
Import-Module Microsoft.Graph.Identity.DirectoryManagement, Microsoft.Graph.Identity.SignIns
Import-Module Microsoft.Graph.Reports
Import-Module ExchangeOnlineManagement
Import-Module ImportExcel
# Optional
Import-Module MicrosoftTeams
Import-Module PnP.PowerShell                       # or: Microsoft.Online.SharePoint.PowerShell
Import-Module Microsoft.Entra
```

5. Connect to each service

Microsoft Graph (users, groups, licences, directory, reports)

```
Connect-MgGraph -NoWelcome -Scopes 'User.ReadWrite.All', 'Group.ReadWrite.All',
    'Directory.Read.All', 'Organization.Read.All', 'AuditLog.Read.All', 'Reports.Read.All',
    'Policy.Read.All'
Get-MgContext | Format-List Account, TenantId, Scopes        # who and what you are connected as
Find-MgGraphCommand -Command Get-MgUser | Select-Object -First 1 -ExpandProperty Permissions
```

*Ask only for the scopes you need; read-only work needs only the *.Read.All scopes. Find-MgGraphCommand shows which permissions a cmdlet requires.*

Exchange Online and Security & Compliance (Purview)

```
Connect-ExchangeOnline -UserPrincipalName admin@yourdomain.com -ShowBanner:$false
Connect-IPPSSession    -UserPrincipalName admin@yourdomain.com     # Purview / compliance
Get-ConnectionInformation | Format-Table Name, UserPrincipalName, TokenStatus
```

Teams, SharePoint and Entra

```
Connect-MicrosoftTeams
Connect-SPOService -Url https://yourtenant-admin.sharepoint.com          # classic SPO module
Connect-PnPOnline  -Url https://yourtenant-admin.sharepoint.com -Interactive -ClientId <appId>
Connect-Entra -Scopes 'User.Read.All', 'Group.Read.All'
```

PnP.PowerShell needs your own Entra app registration (its shared app was retired). Create one once with: Register-PnPEntraIDAppForInteractiveLogin -ApplicationName 'PnP PowerShell' -Tenant yourtenant.onmicrosoft.com -Interactive

Unattended / scheduled scripts (certificate, no prompts)

```
Connect-MgGraph -ClientId <appId> -TenantId <tenantId> -CertificateThumbprint <thumbprint>
Connect-ExchangeOnline -AppId <appId> -CertificateThumbprint <thumbprint> `
    -Organization yourtenant.onmicrosoft.com
Connect-MgGraph -Identity            # Azure Automation / Azure VM managed identity
```

6. Verify and first queries

```
Get-Module Microsoft.Graph*, ExchangeOnlineManagement, MicrosoftTeams |
    Format-Table Name, Version
Get-MgUser -All -Property DisplayName, UserPrincipalName, AccountEnabled |
    Select-Object DisplayName, UserPrincipalName, AccountEnabled
Get-MgSubscribedSku | Select-Object SkuPartNumber, ConsumedUnits,
    @{ n = 'Purchased'; e = { $_.PrepaidUnits.Enabled } }
Invoke-MgGraphRequest -Method GET -Uri 'v1.0/users?$top=5&$select=displayName,mail'
Get-EXOMailbox -ResultSize 10 | Export-Excel .\mailboxes.xlsx -AutoSize -TableName Mailboxes
```

7. Disconnect and remove retired modules

```
Disconnect-MgGraph
Disconnect-ExchangeOnline -Confirm:$false
Disconnect-MicrosoftTeams
Disconnect-PnPOnline
Uninstall-Module MSOnline, AzureAD, AzureADPreview -AllVersions -ErrorAction SilentlyContinue
```

Retired / legacy module	Replace with
MSOnline (Connect-MsolService)	Microsoft.Graph (Get-MgUser, Set-MgUserLicense...) or Microsoft.Entra
AzureAD / AzureADPreview (Connect-AzureAD)	Microsoft.Graph or Microsoft.Entra (similar cmdlet names)
SharePointPnPPowerShellOnline	PnP.PowerShell
Skype for Business Online connector	MicrosoftTeams
Exchange Remote PowerShell (New-PSSession to outlook.office365.com)	ExchangeOnlineManagement v3 (REST based)

8. Troubleshooting

Symptom	Fix
'Could not load file or assembly' when importing Graph	Graph sub-modules are at different versions. Update-Module Microsoft.Graph* so they match, then open a new PowerShell window.
MSAL / sign-in error when using Graph and Exchange together	Both modules ship MSAL. Start a new window and connect to one service at a time, or connect Graph first.
'Function capacity 4096 has been exceeded'	Windows PowerShell 5.1 limit. Use PowerShell 7, or import only the Graph sub-modules you need.
Install-Module: 'Unable to find repository PSGallery'	Register-PSRepository -Default. On 5.1 also set TLS 1.2 (section 1).
Install-Module fails behind a proxy	Allow www.powershellgallery.com (and its CDN) through the proxy, or ask IT for an internal repository.
Insufficient privileges / 403 from Graph	Reconnect with the needed -Scopes; an admin may need to grant consent for the tenant.
Script will not run: 'not digitally signed'	Set-ExecutionPolicy RemoteSigned -Scope CurrentUser, then Unblock-File <script>.

9. All of the above in one command

The repository script **scripts/Initialize-M365AdminModules.ps1** installs, updates, imports and optionally connects in one step, and warns about retired or mismatched modules:

```
.\Initialize-M365AdminModules.ps1                        # Core set: install if missing + load
.\Initialize-M365AdminModules.ps1 -Profile Full -Update   # everything, updated
.\Initialize-M365AdminModules.ps1 -Connect -UserPrincipalName admin@yourdomain.com
.\Initialize-M365AdminModules.ps1 -WhatIf                 # preview, change nothing
```